

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Business Insights & Recommendations

Python • SQL • Power BI

End-to-end retail analytics case study

Prepared from the project dashboard and SQL analysis

3.9K

Customers

\$59.76

Avg. Purchase

3.75

Avg. Rating

Business Problem & Objective

02

Turning customer shopping data into actionable retail decisions

Problem Statement

A leading retail company wants to better understand customer shopping behavior to improve sales, customer satisfaction and long-term loyalty. Management needs to understand how demographics, categories, discounts, reviews and payment or delivery preferences influence purchasing decisions.

Overarching business question

How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?

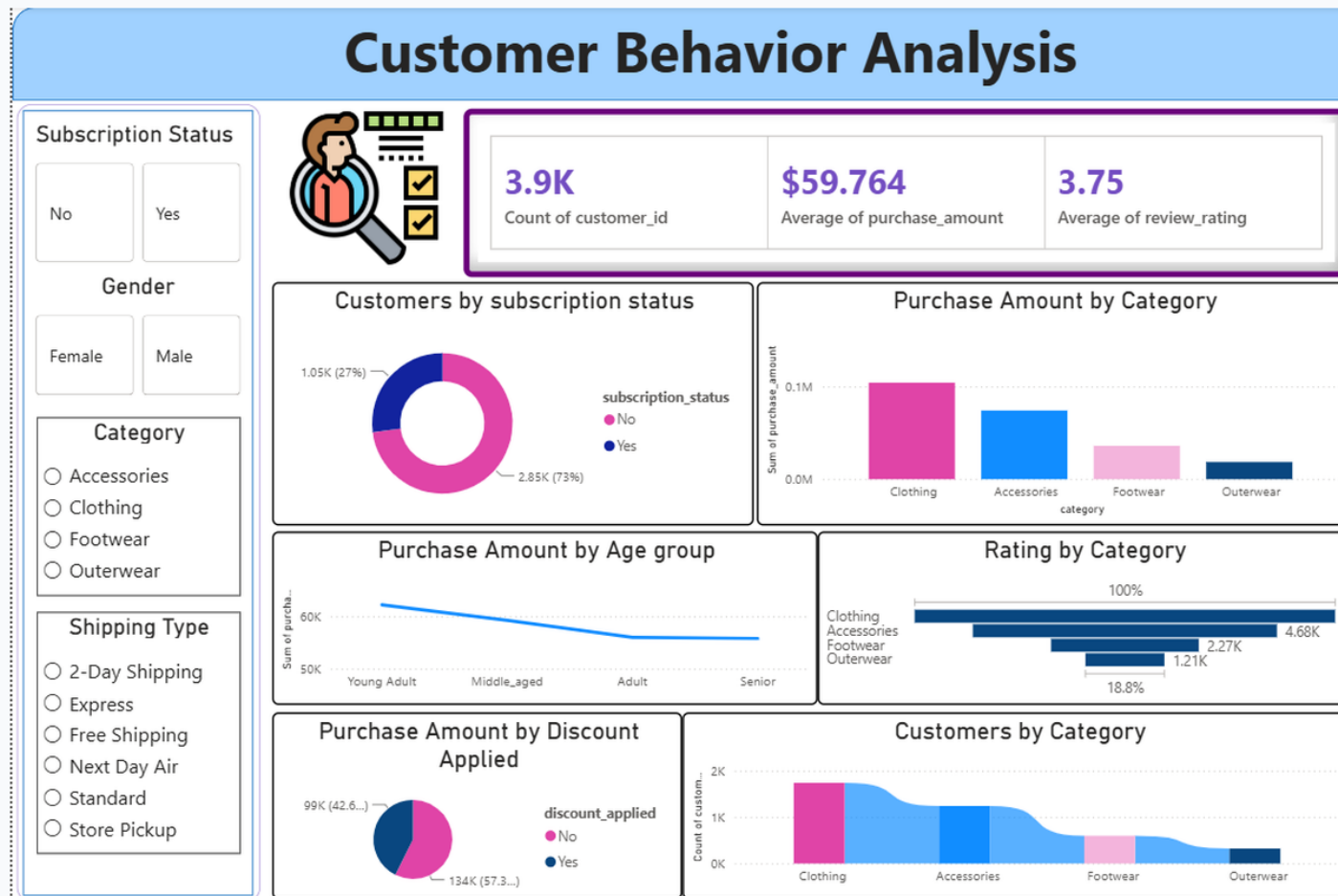
Analytics Objectives

- Profile customers by demographics and purchasing behavior.
- Identify categories and customer groups contributing most to sales.
- Understand discount and subscription behavior.
- Use SQL to answer targeted business questions.
- Communicate KPIs and patterns through an interactive Power BI dashboard.
- Translate findings into practical marketing, product and retention actions.

Executive Dashboard Snapshot

03

Current KPI and category-level view from the Power BI dashboard



Customer & Engagement Insights

04

What the dashboard indicates about the customer base

3.9K

Customers

The dashboard reports approximately 3.9K customers.

\$59.76

Average purchase

Average purchase amount is approximately \$59.76.

3.75

Average rating

Overall average review rating is 3.75.

27%

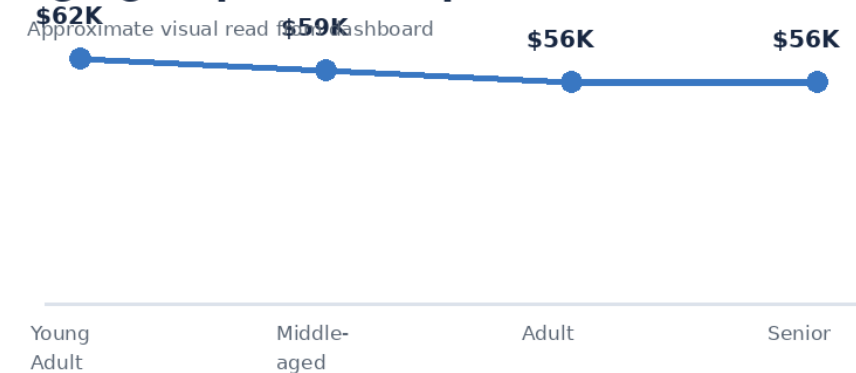
Subscribed

About 27% are subscribed, while 73% are not.

Customer engagement signals

- Subscription penetration is relatively low, creating a clear opportunity to convert engaged non-subscribers.
- The large non-subscriber share suggests retention and loyalty programs could be an important growth lever.
- Young adults contribute the highest purchase amount among the displayed age groups, followed by middle-aged customers.

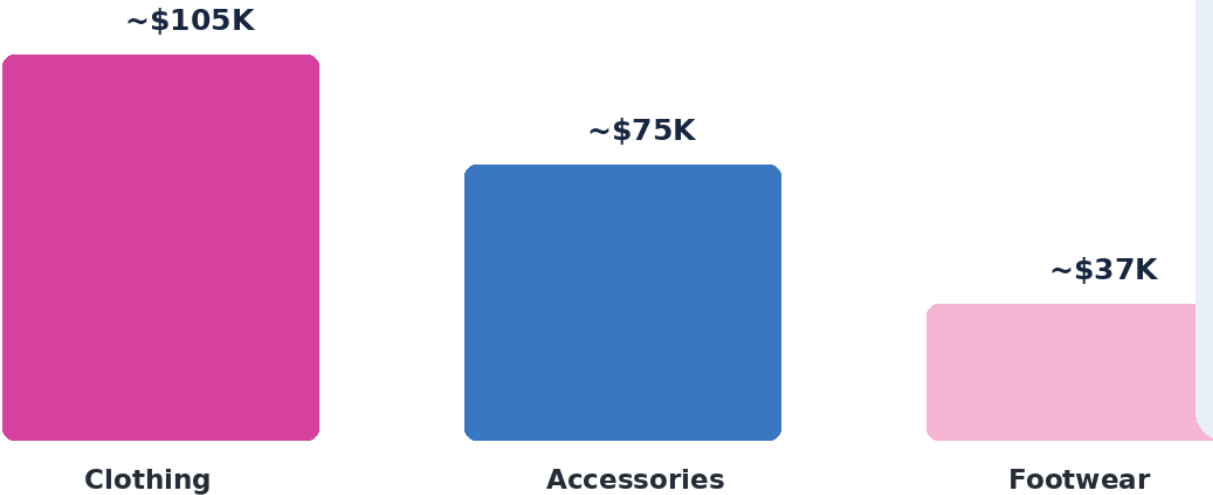
Age-group revenue pattern



Product & Sales Performance

Category-level findings from the dashboard

Purchase Amount by Category



Business interpretation

- Clothing is the strongest category by purchase amount and customer volume.
- Accessories is the second-largest contributor and represents a strong cross-sell opportunity.
- Footwear and Outerwear contribute less revenue and may need targeted promotions or assortment review.
- Category-level performance should guide inventory, merchandising and marketing allocation.

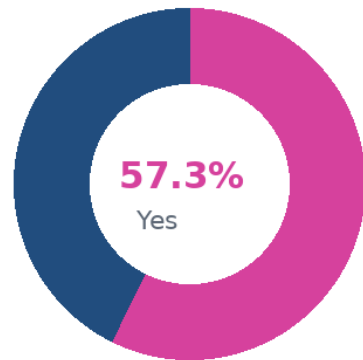
Values are approximate visual readings from the dashboard; use the underlying dataset for exact reporting.

Discount & Subscription Insights

06

Opportunities to improve conversion while protecting profitability

Purchase Amount by Discount



~\$134K

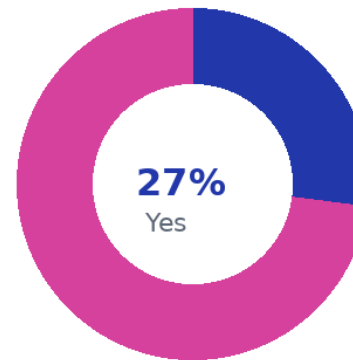
Purchase amount with discount

~\$99K

Purchase amount without discount

More than half of purchase value is associated with discounted transactions in the dashboard.

Subscription Opportunity



~1.05K

Subscribed customers

~2.85K

Non-subscribed customers

Key action:

Test targeted loyalty/subscription offers for engaged non-subscribers.

SQL Analysis & Business Questions

07

How the SQL layer supports deeper decision-making

1. Revenue by gender

Compare male vs. female revenue to understand demographic contribution.

2. Discounted high spenders

Find customers using discounts who still spend above the average purchase amount.

3. Top-rated products

Identify the five products with the highest average review ratings.

4. Shipping behavior

Compare average purchase amounts for Standard vs. Express shipping.

5. Subscription economics

Compare customer count, average spend and total revenue for subscribers vs. non-subscribers.

6. Discount-heavy products

Find the five products with the highest share of discounted purchases.

7. Repeat buyers & subscription

Check whether customers with more than five previous purchases are more likely to subscribe.

8. Revenue by age group

Rank age groups by total revenue contribution.

These SQL questions connect operational data to customer, product, loyalty and pricing decisions.

Business Recommendations

08

Prioritized actions derived from the dashboard and analytical objectives

01 Grow subscription adoption

Use targeted offers, loyalty benefits and personalized messaging to convert high-engagement non-subscribers.

02 Protect discount ROI

Measure incremental sales and margin from discounts; focus promotions on products/customers where discounts drive profitable demand.

03 Prioritize category leaders

Maintain strong availability and marketing for Clothing and Accessories while testing targeted strategies for lower-volume categories.

04 Personalize customer campaigns

Use age, gender, purchase frequency, previous purchases and category preferences for targeted campaigns.

05 Use ratings in merchandising

Combine product ratings with sales performance to identify products that are both popular and well received.

06 Improve repeat-purchase strategy

Use SQL-based repeat-buyer analysis to design retention journeys and evaluate whether loyalty programs increase subscription conversion.